

PROJECT REPORT

Campus Lost and Found System

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Submitted to: Prof. Manorama Chouhan

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Cover Page

Project Title:

Campus Lost and Found System

Project Goal:

- To make it easier for students, faculty and staff to report, search, match and claim lost or found items on campus.
- To apply important Java concepts to a practical real-world problem.

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Date of Submission: 18 September 2026

Project Introduction

A college campus consists of a large number of students, faculty members, staff members and visitors. As a result, personal belongings such as identity cards, books, calculators, wallets, keys, bags, electronic devices and other items may frequently be lost or found.

In many situations, information about these items is communicated through informal channels such as student groups, personal messages or security desks. This can make it difficult to maintain an organized record of all lost and found items. It can also become difficult for a person who has lost an item to determine whether someone else has found it.

The **Campus Lost and Found System** is a command line based Java application developed to provide a structured approach to this problem.

The application allows users to register and log in, report lost and found items, search existing reports, identify possible matches and submit claims for found items.

The project also provides an opportunity to apply concepts from the **CSE2006 – Programming in Java** course in a practical application. These include Object-Oriented Programming, inheritance, polymorphism, collections, exception handling, multithreading, synchronization, file handling and JDBC database connectivity.

The matching component uses a **rule based approach**. Lost and found reports are compared using category, location, date proximity and description similarity to calculate a possible match score.

Problem Statement

Managing lost and found items on a college campus can become difficult when information is scattered across different communication channels.

For example, a student may report a lost calculator to friends or a student group, while another student may find the same calculator and report it separately. Without a centralized system, there may be no systematic way to connect the two reports.

The major problems are:

- Lack of a centralized record of lost and found items.
- Difficulty searching through multiple reports.
- Difficulty identifying potentially matching lost and found items.
- No structured process for claiming found items.
- Difficulty managing the status of reports and claims.
- Possibility of incomplete or duplicated information.

The proposed **Campus Lost and Found System** addresses these problems through a centralized command-line application that provides reporting, searching, matching, and claim-management functionality.

Functional Requirements

Functional requirements describe the operations that the system must perform.

FR1 – User Registration

The system shall allow a new user to register by providing the required information.

The application supports different user types:

- Student
- Faculty
- Staff

FR2 – User Login

The system shall allow registered users to log in using their credentials.

Invalid login information should result in an appropriate error message.

FR3 – Report Lost Item

A logged-in user shall be able to report an item that has been lost.

The report contains relevant information such as:

- Item name
- Category
- Location
- Date
- Description
- Reporter information

FR4 – Report Found Item

A user shall be able to report an item that has been found on campus.

The found report contains information that can later be compared with lost-item reports.

FR5 – Manage Reports

Users shall be able to view and manage their reports.

The system provides operations for relevant report management, including updating, cancelling, or deleting reports where applicable.

FR6 – Search Items

The system shall allow users to search existing lost and found records.

Search can be performed using relevant keywords and available item information.

FR7 – Filter Items

The system shall support filtering of item records using criteria such as:

- Category
- Location
- Date
- Other available item attributes

Multiple criteria can also be used during searching.

FR8 – Find Possible Matches

The system shall compare lost and found reports and calculate a possible match score.

The matching mechanism considers:

Matching Factor	Maximum Score
Category	25
Location	25
Date Proximity	20
Description Similarity	30
Total	100

FR9 – Claim Found Item

A user shall be able to submit a claim for an appropriate found item.

FR10 – Manage Claims

The system shall maintain the status of claims.

The claim workflow includes statuses such as:

- Pending
- Approved
- Rejected

FR11 – File Export

- The system shall provide file-based export functionality for relevant application information.

FR12 – Database Connectivity

The system shall support database connectivity using **JDBC** and MySQL.

Non-Functional Requirements

1 Usability

The application should be easy to operate through a menu-driven command-line interface.

The system provides numbered options and prompts so that users can navigate the application without requiring a graphical interface.

2. Reliability

The application should handle invalid input and application-specific errors without unexpectedly terminating.

Input validation and exception handling are used to improve reliability.

3. Maintainability

The project is divided into multiple packages and classes based on their responsibilities.

This makes the code easier to understand, modify, and extend.

4. Performance

The system uses Java collections such as ArrayList and HashMap to manage and search application data.

Background matching tasks can also be executed using multithreading.

5. Portability

The application is designed to run through the command line on systems supporting the required Java and Maven environment.

6. Security

JDBC database operations use prepared statements where applicable.

Input validation is also performed before processing user-provided information.

7. Scalability

The modular design allows additional functionality to be introduced in the future, such as a web interface, advanced matching, or additional notification mechanisms.

Project Structure



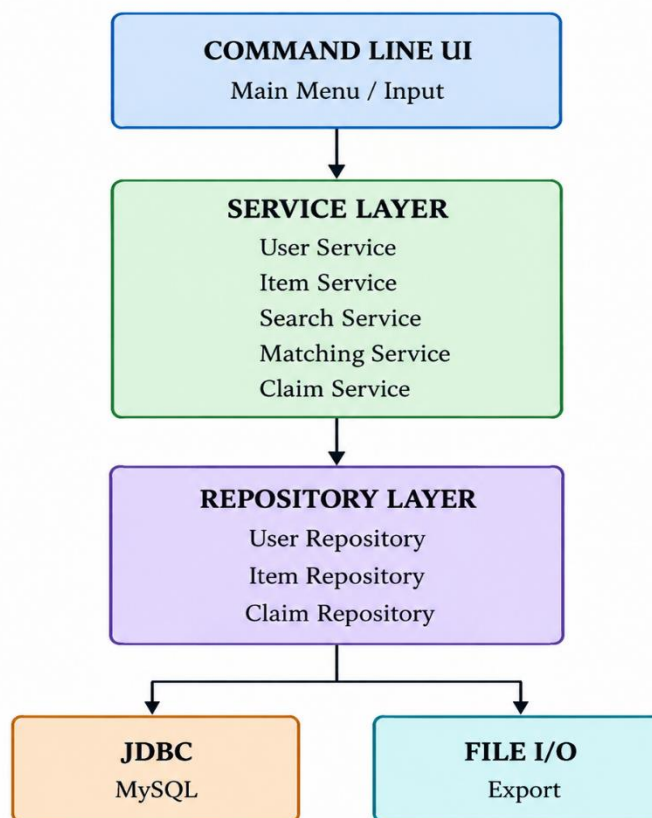
System Architecture

The application follows a modular layered architecture.

The main layers are:

1. Command Line Interface
2. Service Layer
3. Repository Layer
4. Database / File Storage Layer

Architecture



Architecture Flow

The user interacts with the system through the command line interface.

The selected operation is passed to the appropriate service. The service performs the required business logic and communicates with repositories for data operations.

Repositories interact with the database through JDBC where database persistence is used. File-related operations are handled through the file I/O components.

Design Diagrams

1. Use Case Diagram

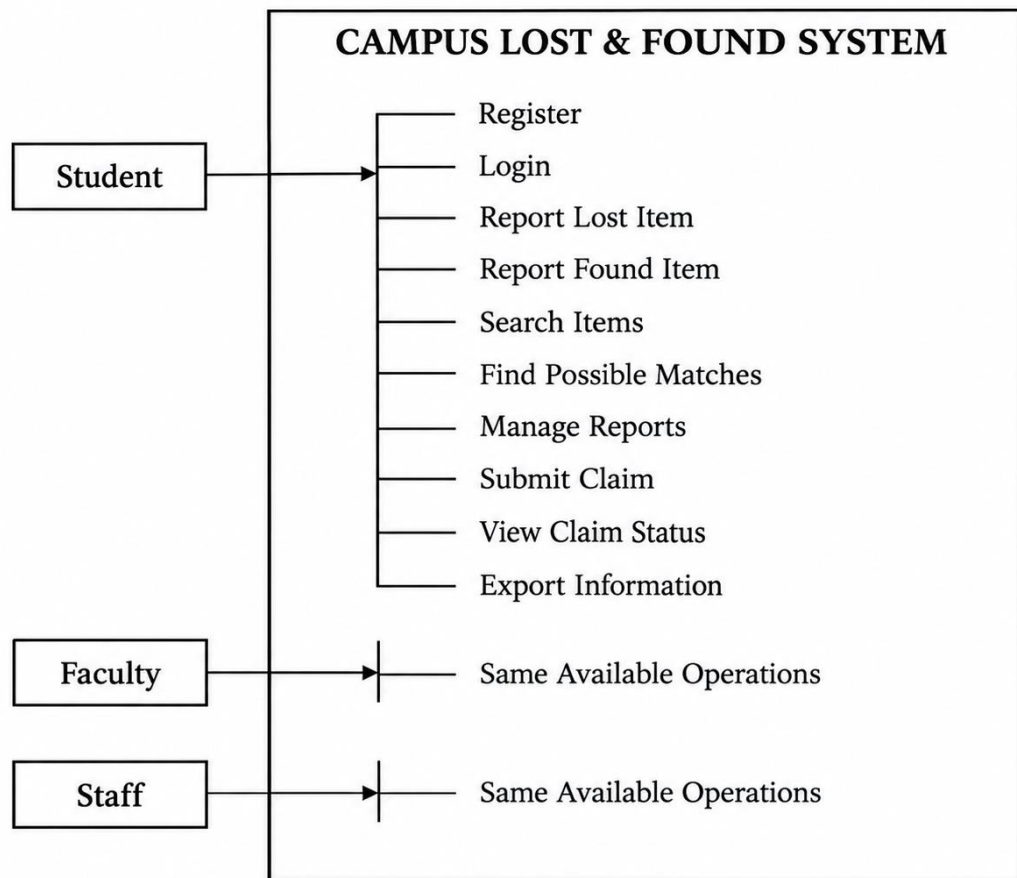
The major actors are:

- Student
- Faculty
- Staff

The major use cases are:

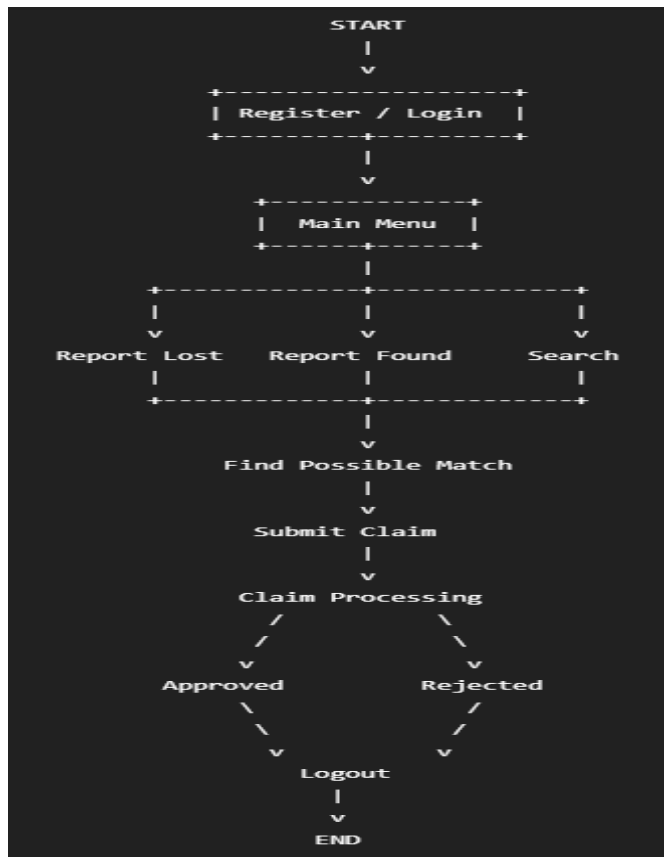
- Register
- Login
- Report Lost Item
- Report Found Item
- Search Items
- Find Possible Matches
- Manage Reports
- Submit Claim
- Manage Claims
- Export Information

2. User Case Representation

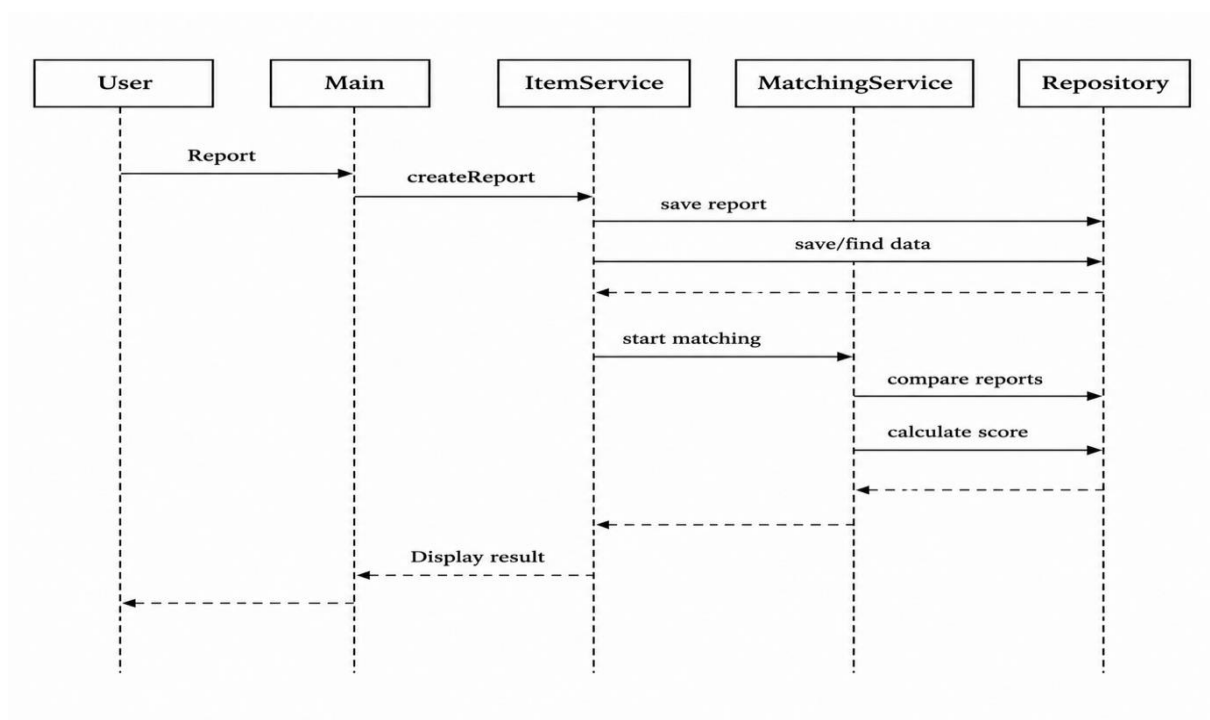


3. Workflow Diagram

The overall workflow of the application is:



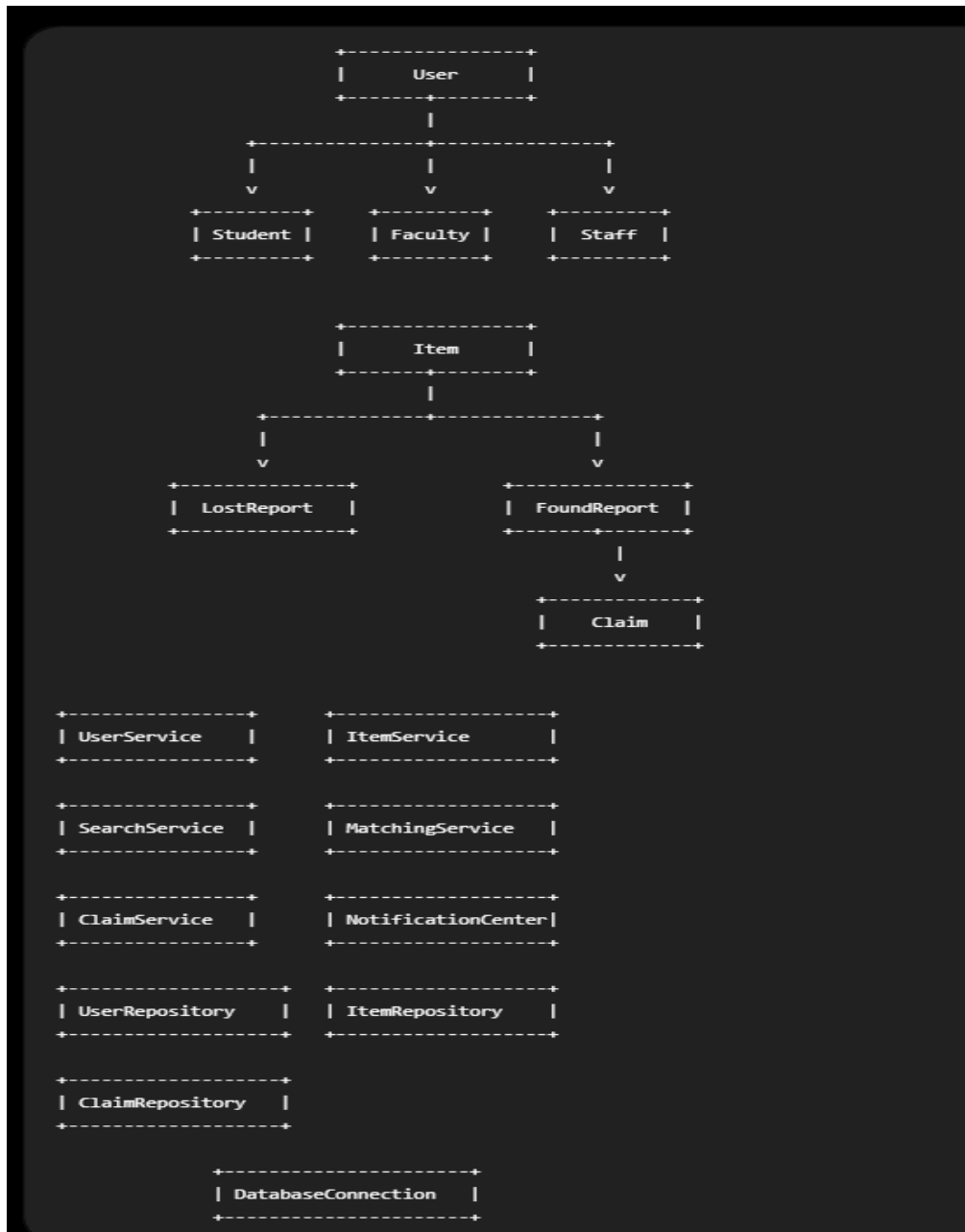
4. Sequence Diagram



This illustrates the interaction between the user interface, service layer, matching component, and repository.

5. Class / Component Diagram

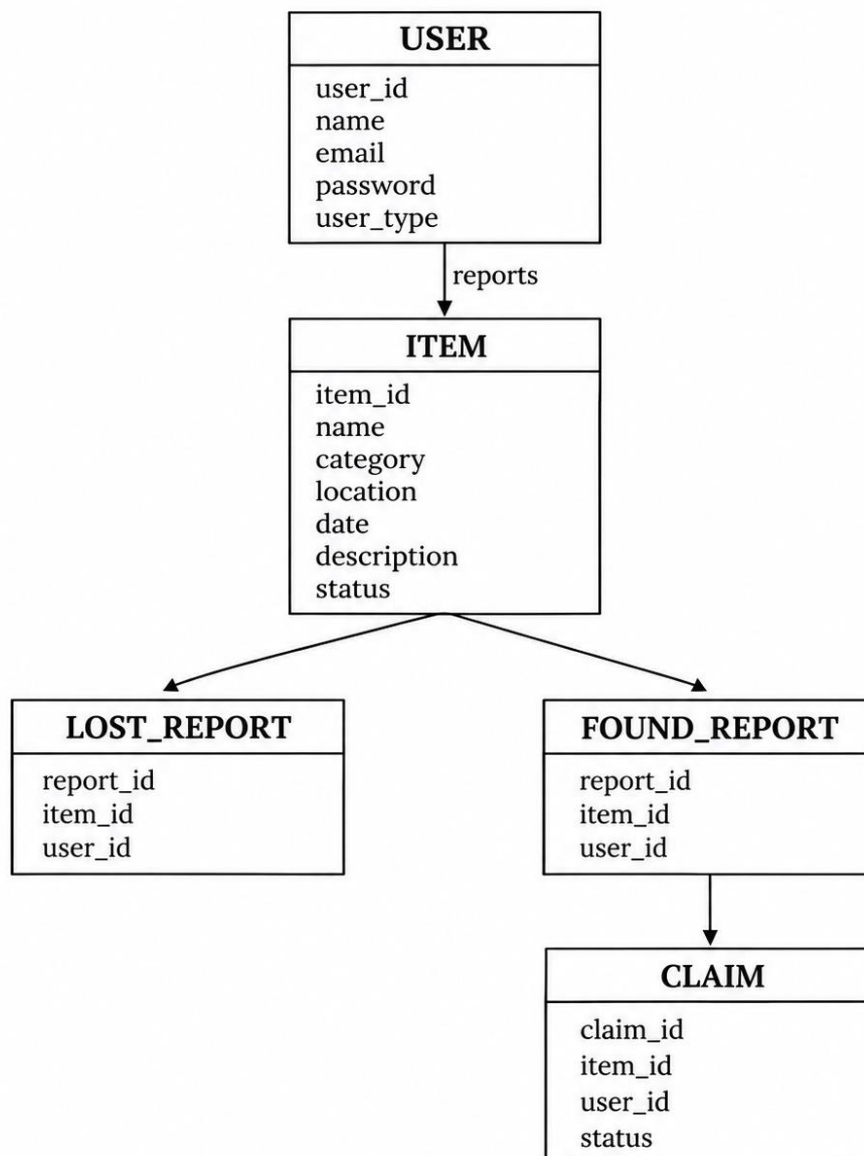
The major classes can be represented as follows:



Main Relationships

- Student, Faculty, and Staff inherit from User.
- LostReport and FoundReport are associated with item information.
- Claim represents a claim against a found item.
- Services contain application logic.
- Repositories handle data access.
- DatabaseConnection manages JDBC database connectivity.

06. ER Diagram



The exact attributes and relationships should correspond to the final schema.sql included in the GitHub repository.

Design Decisions and Rationale

Decision	Rationale
Command Line Interface	Chosen to keep the application simple and directly executable from the terminal.
Object Oriented Design	Used to represent users, items, reports, and claims through Java classes.
Inheritance	Student, Faculty, and Staff inherit common properties from User.
Service Repository Structure	Separates business logic from data access, making the project modular.
Rule based Matching	Uses category, location, date, and description instead of Machine Learning.
JDBC	Used for database connectivity as required by the Java syllabus.
Collections	ArrayList and HashMap are used for storing and searching records.
Multithreading	Used for background matching-related tasks.

Implementation Details:

- Developed using **Java and Maven**.
- Provides user registration and login.
- Allows users to report lost and found items.
- Supports searching and filtering of reports.
- Uses a **100-point rule-based matching system**.
- Provides claim submission and claim-status management.
- Uses custom exceptions and input validation.
- Uses Java File I/O for exporting information.
- Uses JDBC for MySQL database connectivity.

- Uses Runnable and synchronization for multithreading related functionality.
- Code is organized into model, service, repository, database, exception, enums and util packages.

Program Execution and Results:

- The project successfully meets all functional requirements.
- A user is able to interact with the registration/login interface and all submitted data is validated in real-time.

Here, I am pasting my all the results of my programs when executed:

1. Main Menu

```
=====
                CAMPUS LOST AND FOUND SYSTEM – VIT BHOPAL
                Course: CSE2006 Programming in Java
=====

=====
                        MAIN MENU
=====
1. Login
2. Register New Account
3. Search & Browse Campus Items (Public)
4. Find Possible Matches (Public Campus Overview)
5. Exit
=====
Enter your choice (1-5): 2
```

2. Registration

```

=====
Enter your choice (1-5): 2

--- NEW USER REGISTRATION ---
Enter full name: Utkarsh
Enter campus email: utkarsh.25bai11231@vitbhopal.ac.in
Enter password (min 4 characters): 1234
Enter phone number: 9999999999
Enter department (e.g. SCOPE, SCSE, SENSE, Civil): SCAI

Select Role:
1. Student
2. Faculty
3. Staff
Select (1-3): 1
Enter Registration Number (e.g. 23BCE10001): 25BAI11231
Enter Hostel Block & Room (or type 'Day Scholar'): Block 04B Room G010
[Database Notice] MySQL server is not reachable (Communications link failure)

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the server.).
[Database Notice] Operating in in-memory fallback mode so all features work seamlessly.

[+] Registration successful! You are now logged in.

```

```

[+] Registration successful! You are now logged in.

```

```

-----
User ID       : 4
Name          : Utkarsh
Email         : utkarsh.25bai11231@vitbhopal.ac.in
Phone        : 9999999999
Role          : STUDENT
Department    : SCAI
Role Details  : Reg No: 25BAI11231 | Hostel/Room: Block 04B Room G010
-----

```

3. User Dashboard

```

=====
                        USER DASHBOARD
Logged in as: Utkarsh | Role: STUDENT (25BAI11231)
=====
1. Report Lost Item
2. Report Found Item
3. Search Items (Keyword / Multi-Criteria Filter)
4. Find Possible Matches (Smart Rule-Based Engine)
5. Manage My Reports (View / Update / Cancel / Delete)
6. Claim Management (Submit Claim / Review Claims)
7. Export Reports to File
8. View My Profile
9. Logout
=====
Enter your choice (1-9): 1

```

4. Reporting a Lost Item

```
=====
Enter your choice (1-9): 1

--- REPORT A LOST ITEM ---
Enter Item Name / Title: Book
Select Item Category:
1. ELECTRONICS
2. ID_CARD
3. BOOKS_STATIONERY
4. BAGS
5. KEYS
6. CLOTHING
7. BOTTLE
8. OTHER
Enter category number (1-8): 3
Enter Location where lost (e.g. Library 2nd Floor, AB-1 Lab 102): AB1 3rd Floor
Enter Date lost (YYYY-MM-DD) [Press Enter for today (2026-09-18)]: 2026-09-17
Enter Detailed Description (colors, unique markings, contents): Morris Mano Book for DLCA
Enter Reward amount in INR (0 for none): Rs 50
(!) Invalid decimal number.
Enter Reward amount in INR (0 for none): 50
Does it have unique identifiable markings/serial numbers (yes/no)? yes

[+] Lost item report created successfully! Report ID: #5
=====
```

5. Re-jump to User Dashboard

```
=====
                        USER DASHBOARD
Logged in as: Utkarsh | Role: STUDENT (25BAI11231)
=====
1. Report Lost Item
2. Report Found Item
3. Search Items (Keyword / Multi-Criteria Filter)
4. Find Possible Matches (Smart Rule-Based Engine)
5. Manage My Reports (View / Update / Cancel / Delete)
6. Claim Management (Submit Claim / Review Claims)
7. Export Reports to File
8. View My Profile
9. Logout
=====
Enter your choice (1-9): 3
```

6. Search and Filter Items

```
--- SEARCH & FILTER ITEMS ---
1. View All Campus Reports
2. Simple Keyword Search (Title, Description, or Location)
3. Quick Category Filter (Using HashMap Category Index)
4. Multi-Criteria Search (Category + Location + Date + Status + Keyword)
Select option (1-4): 2
Enter keyword to search: Book
-----
ID | TYPE | TITLE | CATEGORY | STATUS | LOCATION | DATE
-----
5 | LOST | Book | BOOKS_STATIONERY | OPEN | AB1 3rd Floor | 2026-09-17
-----
```

7. Re-jump to User Dashboard

```
=====
                        USER DASHBOARD
Logged in as: Utkarsh | Role: STUDENT (25BAI11231)
=====
1. Report Lost Item
2. Report Found Item
3. Search Items (Keyword / Multi-Criteria Filter)
4. Find Possible Matches (Smart Rule-Based Engine)
5. Manage My Reports (View / Update / Cancel / Delete)
6. Claim Management (Submit Claim / Review Claims)
7. Export Reports to File
8. View My Profile
9. Logout
=====
Enter your choice (1-9): 2
```

8. Report a Found Item

```
--- REPORT A FOUND ITEM ---
Enter Item Name / Title: Book
Select Item Category:
1. ELECTRONICS
2. ID_CARD
3. BOOKS_STATIONERY
4. BAGS
5. KEYS
6. CLOTHING
7. BOTTLE
8. OTHER
Enter category number (1-8): 3
Enter Location where found: AB 1 Third Floor\
Enter Date found (YYYY-MM-DD) [Press Enter for today (2026-09-18)]:
Enter Detailed Description: The book has been found in room 331
Where is the item deposited / kept (e.g. Security Main Gate, AB-1 Proctor Office)? To Library

[+] Found item report created successfully! Report ID: #6
```

9. Re-jump to User Dashboard

```

=====
                        USER DASHBOARD
Logged in as: Utkarsh | Role: STUDENT (25BAI11231)
=====
1. Report Lost Item
2. Report Found Item
3. Search Items (Keyword / Multi-Criteria Filter)
4. Find Possible Matches (Smart Rule-Based Engine)
5. Manage My Reports (View / Update / Cancel / Delete)
6. Claim Management (Submit Claim / Review Claims)
7. Export Reports to File
8. View My Profile
9. Logout
=====
Enter your choice (1-9):

```

10. Item Matching and Claiming

```

=====
Enter your choice (1-9):
>>> [BACKGROUND ALERT] Match engine identified 1 potential match(es) for your item 'Book' (#6).
> 6

--- CLAIM MANAGEMENT ---
1. Submit a Claim for a Found Item
2. View My Submitted Claims
3. Review Claims for Items I Found (Approve / Reject)
Select choice (1-3): 1

Available Found Items:
=====
ID      | TYPE  | TITLE                                | CATEGORY      | STATUS  | LOCATION          | DATE
=====
2       | FOUND | Navy Blue HP Laptop Bag             | BAGS          | OPEN    | Central Library... | 2026-09-15
4       | FOUND | Red Lanyard Student I...            | ID_CARD       | OPEN    | Food Street Caf... | 2026-09-16
6       | FOUND | Book                                | BOOKS_STATIONERY | OPEN    | AB 1 Third Floor\  | 2026-09-18
=====
Enter Found Item ID you want to claim: 6
Enter your Lost Item ID (or 0 if you did not report one): 6
Enter Proof of Ownership (e.g. lock screen PIN, invoice, serial number, stickers): Morris Mano Book for DLCA
(!) Claim submission failed: Invalid claim: You cannot claim an item that you reported finding.

```

11. Match Engine and Re-jump to User Dashboard

```

-----
? PENDING BACKGROUND NOTIFICATIONS:
  * Match engine identified 1 potential match(es) for your item 'Book' (#6)
-----

=====
                        USER DASHBOARD
Logged in as: Utkarsh | Role: STUDENT (25BAI11231)
=====
1. Report Lost Item
2. Report Found Item
3. Search Items (Keyword / Multi-Criteria Filter)
4. Find Possible Matches (Smart Rule-Based Engine)
5. Manage My Reports (View / Update / Cancel / Delete)
6. Claim Management (Submit Claim / Review Claims)
7. Export Reports to File
8. View My Profile
9. Logout
=====
Enter your choice (1-9): 8

```

12. Logged in User Profile

```
--- LOGGED-IN USER PROFILE ---  
-----  
User ID       : 4  
Name          : Utkarsh  
Email         : utkarsh.25bai11231@vitbhopal.ac.in  
Phone        : 9999999999  
Role          : STUDENT  
Department    : SCAI  
Role Details  : Reg No: 25BAI11231 | Hostel/Room: Block 04B Room G010  
-----  
=====
```

13. Logged in Dashboard

```
=====
```

USER DASHBOARD

Logged in as: Utkarsh | Role: STUDENT (25BAI11231)

```
=====
```

1. Report Lost Item
2. Report Found Item
3. Search Items (Keyword / Multi-Criteria Filter)
4. Find Possible Matches (Smart Rule-Based Engine)
5. Manage My Reports (View / Update / Cancel / Delete)
6. Claim Management (Submit Claim / Review Claims)
7. Export Reports to File
8. View My Profile
9. Logout

```
=====
```

Enter your choice (1-9): 9

14. Logging out and Main Menu

```
=====
```

Enter your choice (1-9): 9

[+] Logged out Utkarsh successfully.

```
=====
```

MAIN MENU

```
=====
```

1. Login
2. Register New Account
3. Search & Browse Campus Items (Public)
4. Find Possible Matches (Public Campus Overview)
5. Exit

```
=====
```

Enter your choice (1-5): 5

Testing Approach:

Testing was performed using functional and integration testing. It:

- Tested user registration and login.
- Tested valid and invalid inputs.
- Tested lost and found item reporting.
- Tested searching and filtering.
- Tested possible-match calculation.
- Tested claim submission and status changes.
- Tested file export.
- Tested exception handling.
- Tested complete workflow from reporting to claiming.
- Tested project compilation using Maven.

Challenges Faced:

I faced the following barriers/ challenges:

- Designing an appropriate Java class structure.
- Managing different user types using inheritance.
- Connecting different modules into one workflow.
- Designing a rule-based matching mechanism.
- Handling invalid command-line inputs.
- Managing JDBC database connectivity.
- Implementing background tasks using multithreading.
- Maintaining modular and organized project code.

Learnings and Key Takeaways

These types of simple projects for beginners provides them valuable experience in several key areas of application development.

I gained these takeaways:

- Gained practical understanding of **Object-Oriented Programming**.
- Learned to use inheritance and polymorphism in a real application.

- Applied Java Collections such as ArrayList and HashMap.
- Improved understanding of exception handling and validation.
- Learned practical implementation of multithreading and synchronization.
- Gained experience with Java File I/O.
- Learned how Java applications communicate with databases using JDBC.
- Understood the importance of modular project structure and documentation.
- Gained experience with Maven, Git, and GitHub.

Potential Future Enhancements:

To expand and enhance the functionality and professionalize the application, the following features could be added:

- Develop a web-based interface.
- Develop a mobile application.
- Add email/SMS notifications.
- Integrate institutional login.
- Add image based item matching.
- Extend the rule based system with Machine Learning.
- Deploy the database to the cloud.
- Add an administrative dashboard.
- Provide advanced reports and analytics.

References:

- Java Standard Library Documentation (Javadoc, JDK, Maven).
- GeeksforGeeks Java Programming Tutorials.
- YouTube lectures for basic understandings of Java programming. (Code With Harry).
- VITyarthi

THANK YOU!